DAY 5 | POST-READ

APIs and FastAPI Core: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
HTTP request and response	Method + URL + headers + optional body; status code + JSON back.
REST	Plural nouns, methods as verbs, stateless, consistent.
Status codes	200, 201, 204, 400/422, 401/403, 404, 5xx.
FastAPI	app = FastAPI(); @app.get(...) decorators; uvicorn main:app --reload; /docs.
Parameters	Path identifies, query filters, body carries data. Query() adds rules.
APIRouter	Group routes by feature with prefix and tags, then include_router.
Pydantic	BaseModel with typed fields; Field() for rules; invalid data returns 422.
PATCH pattern	model_dump(exclude_unset=True) then model_copy(update=...).
Codex review (NEW v4)	Small prompt, read every diff, prove it with TestClient tests.

Part B: Going beyond today

1. Idempotency explained

Pressing "pay" twice should not charge twice. PUT and DELETE are idempotent; POST is not, so real payment APIs accept an idempotency key header to detect duplicates.

2. Custom validators

```
from pydantic import BaseModel, field_validator

class BookCreate(BaseModel):
    isbn: str

    @field_validator("isbn")
    @classmethod
    def check_isbn(cls, v):
        if len(v.replace("-", "")) != 13:
            raise ValueError("ISBN must have 13 digits")
        return v
```

3. Enums for fixed choices

class Genre(str, Enum) with a fixed list of values: only listed values are accepted, and /docs shows a dropdown.

4. API versioning

Prefix routes with /v1 so you can change things later without breaking existing apps.

5. Tests as a safety net for AI code

Write the test before you ask Codex for the endpoint. If the generated code passes a test you wrote yourself, you have real evidence. If you let Codex write both, read the tests extra carefully: a test that cannot fail proves nothing.

Part C: Check yourself

- Which method would you use to change only the price of a product?
- What is the difference between 401 and 403?
- How does FastAPI decide whether a parameter is path, query or body?
- Why do we use separate Create and Response models?
- Name two things you check in every Codex diff.

Part D: Practice ideas for the weekend

- Add GET /quotes/random and POST /quotes returning 201; return 404 for unknown ids.
- Add a genre Enum and a validator that rejects titles in all capitals.
- Ask Codex for one endpoint, then write a test that would have caught a mistake in it.

Coming up on Day 6: SQLAlchemy models, relationships, transactions and Swagger docs.

Yoi shuumatsu wo (良い週末を): have a great weekend